A fixed Bob-complement certificate for the R1 source box

1 The complementary Bob region

The same native source box admits a second fixed Gram matrix on $|v| \geq 1/16$, with no restriction on the Alice angle. This certificate uses the entire state quadratic form.

Use the compact testing coordinates

$$(a,b) = \frac{(u,1)}{\sqrt{1+u^2}}, \qquad (c,e) = \frac{(v,1)}{\sqrt{1+v^2}}, \qquad a^2+b^2 = c^2+e^2 = 1, \quad b \geq 0.$$

The circle coordinates include the projective endpoints. For independent real state coordinates W, X, Y, Z, define

$$\begin{aligned} E &= aW - bY, \quad K = aX - bZ, \quad J = eE + cK, \quad U = eZ - cY, \\ E_\lambda &= eW + cX, \qquad L_A = bX + aZ, \qquad M = b(cW - eX) + a(cY - eZ). \end{aligned}$$

For source order $(d, \lambda, n, m, C, A, D, \alpha, \beta)$, the compact quarter-parent is

$$P = \lambda E_\lambda^2 + \beta L_A^2 + nX^2 + mM^2 + CE^2 + AY^2 + (D + d/2)J^2 + (\alpha + d/2)U^2 + \tfrac{d}{2}(L_A^2 - Z^2). \tag{1}$$

For finite angles this is the canonical quarter-parent divided by its positive angular denominators.

Put $q_B = 1/257$, $\tau_B = 257d/2$, and

$$S_B = \begin{pmatrix} C - \tau_B & \tau_B \\ \tau_B & D - \tau_B \end{pmatrix}.$$

Since $J - eE = cK$ and $L_A^2 - Z^2 = X^2 - K^2$, one has the exact identity

$$CE^2 + DJ^2 - \tfrac{d}{2}K^2 = C(cE)^2 + (eE, J)S_B(eE, J)^T + \tau_B(c^2 - q_B)K^2. \tag{2}$$

Consequently P is represented on the ten fixed functions

$$\Phi_B = (E_\lambda, L_A, X, M, Y, U, J, cE, eE, \sqrt{c^2 - 1/257}\, K)^T. \tag{3}$$

Explicitly, let e_i, $0 \leq i \leq 9$, denote the standard basis of $\mathbb{R}^{10}$. Its source-affine Gram matrix is

$$\begin{aligned} Q_B = {} & \operatorname{diag}(\lambda, \beta, n + d/2, m, A, \alpha + d/2, D - 128d, C, C - 257d/2, 257d/2) \\ & + \tfrac{257d}{2}(e_6 e_8^T + e_8 e_6^T), \qquad P = \Phi_B^T Q_B \Phi_B. \end{aligned} \tag{4}$$

This is an identity for every source, rather than an identity after a special choice of auxiliary variables. There are no auxiliaries. For nonnegative source coordinates, $S_B \succeq 0$ is sufficient for $Q_B \succeq 0$: its $(8,6)$ block is S_B plus the positive diagonal contribution $d/2$ on row 6, and all other terms are positive diagonal.

Uniform source admission

Take the native R1 source box

$$\delta = 2^{-10}, \quad 0 < B, T, x = jB \leq \delta, \quad 0 < p < \min(j, 1), \quad 0 < H < 1, \quad n \geq 0, \qquad (d, \lambda, \beta) = (1, 0, 0).$$

Let $J_s = j + 1$, $K_s = j + 1 - p$ and retain the source constants

$$f_0 = 1 - \delta - \delta^3, \qquad R_0 = 1 + 2\delta + 2\delta^2 + 4\delta^3 + \delta^4(1 + 2\delta).$$

The native dictionary gives

$$C = \frac{L}{2HTP}, \qquad D = \frac{(1+Hy)(K_s+Hjy)[1-y((1-H)j-p)]}{2T(1-H)P}, \qquad y = TB,$$

where $P \le R_0 J_s$, $L \ge f_0 K_s$ and $K_s/J_s \ge 1/2$. Since $yj = Tx \le \delta^2$,

$$C \ge \frac{f_0}{4R_0HT}, \qquad D \ge \frac{1-\delta^2}{4R_0T(1-H)}. \tag{5}$$

Here $f_0 < 1-\delta^2$. Setting $k = f_0/(4R_0T)$ therefore gives $C \ge k/H$ and $D \ge k/(1-H)$. Their reciprocal sum satisfies

$$\frac{1}{\eta} := \frac{1}{C} + \frac{1}{D} \le \frac{H}{k} + \frac{1-H}{k} = \frac{1}{k}, \qquad \eta = \frac{CD}{C+D} \ge \frac{f_0}{4R_0T} > \frac{1}{5T} \ge \frac{1024}{5} > \frac{257}{2}.$$

The numerical source comparison is the exact rational inequality

$$\frac{f_0}{4R_0} = \frac{140600049270784}{564050540888577} > \frac{1}{5}.$$

For positive C, D, the matrix $\left(\begin{smallmatrix} C-\tau & \tau \\ \tau & D-\tau \end{smallmatrix}\right)$ is positive definite whenever $0 \le \tau < \eta$, because $\eta < \min(C, D)$ and its determinant is $CD - \tau(C+D)$. Thus $S_B \succ 0$ for every displayed native source.

This argument retains literal $n = 0$ and the zero coordinates $\lambda = \beta = 0$. Arbitrary nonnegative event additions also remain admitted: at $d = 0$, the matrix in (4) is positive semidefinite for each of the eight event rays. No event weight or state coordinate is inverted.

The joined testing domain

For finite Bob angles, $c^2 \ge 1/257$ is exactly $v^2 \ge 1/256$. The functions and identity include both boundary signs, $c = \pm 1/\sqrt{257}$, and the Bob projective endpoints $c = \pm 1, e = 0$. Alice is unrestricted on its compact circle, including $a = \pm 1, b = 0$. All state signs are retained.

Intersecting this matrix constraint with the enlarged cap system $|u| \le 9/256$, $|v| \le 1/16$ gives total positive-semidefinite order

$$274 \text{ (cap)} + 10 \text{ (Bob)} + 9 \text{ (shared source signs)} = 293.$$

It admits the same native sources and event additions. Its testing validity covers the union of those two regions. The remaining strip is

$$|u| > 9/256, \qquad |v| < 1/16,$$

including the Alice projective endpoints within that strip. Admission on this strip and global canonical adequacy are separate questions.

The rational sparse matrix has fifteen upper-triangular records in `bob_complement_order10.coo.json`; indices start at zero and each off-diagonal record is mirrored once. Its exact reconstruction and all nine parent coefficient identities are verified by `check_bob_complement.py`.